* Newtonian Mechanics Review

- Describing motion of "particle" or "point mass"
- Particle $\Rightarrow$ configuration is completely described by position $\vec{r}$

"abstract vector"

- In "coordinate-free" form:

$$\vec{F} = \frac{d}{dt}\Big(\underbrace{m}_{\text{mass}} \frac{d\vec{r}}{dt}\Big) = \frac{d}{dt}(m \underbrace{\vec{v}}_{\text{velocity}}) = \frac{d\vec{p}}{dt}$$

momentum

- Assume constant mass:

$$\vec{F} = \frac{d}{dt}(m\vec{v}) = m\frac{d\vec{v}}{dt} = m\vec{a}$$

acceleration

- Note that we can't do any computation until we chose coordinates
- 2D Cartesian Coordinates

unit basis vectors

$$\vec{r} = r_x \hat{e}_x + r_y \hat{e}_y$$

scalar components

$$= \underbrace{\begin{bmatrix} r_x \\ r_y \end{bmatrix}^T}_{2\times 1 \text{ matrix}} \underbrace{\begin{bmatrix} \hat{e}_x \\ \hat{e}_y \end{bmatrix}}_{\text{"vectrix"}}$$

$$\Rightarrow \vec{v} = \frac{d\vec{r}}{dt} = \dot{r}_x \hat{e}_x + r_x \cancelto{0}{\dot{\hat{e}}_x} + \dot{r}_y \hat{e}_y + r_y \cancelto{0}{\dot{\hat{e}}_y}$$

$$= \dot{r}_x \hat{e}_x + \dot{r}_y \hat{e}_y$$

$$\Rightarrow \vec{a} = \ddot{r}_x \hat{e}_x + \ddot{r}_y \hat{e}_y$$

$$\underbrace{\begin{bmatrix} F_x \\ F_y \end{bmatrix}^T \begin{bmatrix} \hat{e}_x \\ \hat{e}_y \end{bmatrix}}_{\vec{F}} - m \underbrace{\begin{bmatrix} \ddot{r}_x \\ \ddot{r}_y \end{bmatrix}^T \begin{bmatrix} \hat{e}_x \\ \hat{e}_y \end{bmatrix}}_{\vec{a}} = 0$$

$$\Downarrow$$

$$\underbrace{\left(\begin{bmatrix} F_x \\ F_y \end{bmatrix} - m \begin{bmatrix} \ddot{r}_x \\ \ddot{r}_y \end{bmatrix} \right)}^T \underbrace{\begin{bmatrix} \hat{e}_x \\ \hat{e}_y \end{bmatrix}} = 0$$

we can "factor out" the basis vectors

$$\Downarrow$$

$$\underbrace{\begin{bmatrix} F_x \\ F_y \end{bmatrix} = m \begin{bmatrix} \ddot{r}_x \\ \ddot{r}_y \end{bmatrix}}$$

Now we have an expression purely in components (matrices/numbers) that we can compute with.